

BrightLight Tutorials – Data Analytics

SQL Fundamentals

Exercise 3: SQL CASE Statements

Instructions:

1. Write your answers on paper using a pen.
2. For each query, **draw a table** showing the final output (the result set).
3. In your **SELECT** statements, choose **relevant columns to display**, unless specified.
4. After completing all questions, **scan your work into a PDF**.
5. Email your PDF to: roffiwa@brightlighttutorials.co.za
6. **Submission Deadline: 21 April 2025, 23:59**

Questions

1. Table: **products**

product_id	product_name	price	price - category
1	Laptop	1200.00	expensive
2	Phone	800.00	mid-range
3	Keyboard	45.00	Budget
4	Monitor	300.00	Mid-range
5	Mouse	25.00	Budget

Question:

Classify each product by price:

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Expected Output Columns:

- product_name
- price
- price_category

2. Table: orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Question:

Label each order:

- 'High Value' for orders ≥ 1000
- 'Medium Value' for 500–999.99
- 'Low Value' for orders < 500

Expected Output Columns:

- customer_name
- amount
- order_value_category

Customer Name	Amount	Order Value Category
Ethan	1200.00	High Value For order ≥ 1000
Bob	560.00	Medium Value For 500 - 999.99
Charlie	999.99	
Diana	45.50	Low value For
Alice	150.00	Orders < 500

3. Table: employees

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Question:

Categorize employee position:

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

Expected Output Columns:

- emp_name
- department
- salary
- position_level

Emp_Name	Department	Salary	Position-level
John	IT	85000	Senior-IT
Tom	HR	55000	Experienced HR
Sara	HR	60 000	Experienced HR
Mark	IT	75000	Staff.
Lucy	Finance	95000	Staff.

4. Table: **students**

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Question:

Assign a letter grade:

- ≥ 90 : 'A'
- 80–89: 'B'
- 70–79: 'C'
- 60–69: 'D'
- < 60 : 'F'

Expected Output Columns:

- student_name
- score
- grade

Student Name	Score	Grade
Anna	92	A
David	83	B
Ben	76	C
Ella	68	D
Cara	59	F

5. Table: **deliveries**

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Question:

Label delivery performance:

- ≤ 30 mins: 'Fast'
- 31–60 mins: 'On Time'
- 60 mins: 'Late'

Expected Output Columns:

- delivery_id
- delivery_time_minutes
- performance

Delivery-ID	Delivery-Time-minutes	Performance.
3	30	Fast.
1	45	On Time.
4	65	Late
2	80	Late
5	100	Late

6. Table: tickets

ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

Question:

Convert priority to labels:

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

Expected Output Columns:

- issue_type
- priority
- priority_label

Issue - Type	Priority	Priority - Label
Server down	3	High.
Slow System	2	medium
Email Error	2	Medium
Long issue	1	Low
Password Reset	1	Low

7. Table: attendance

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Question:

Calculate attendance % and classify:

- $\geq 90\%$ → 'Excellent'
- 75–89% → 'Good'
- $< 75\%$ → 'Needs Improvement'

Expected Output Columns:

- student_id
- attendance_percentage
- attendance_status

Student-ID	Attendance-Percentage	Attendance-Status
5	100%	Excellent.
3	96%	Excellent.
1	90%	Excellent.
2	60%	Needs-Improvement.
4	50%	Needs-Improvement.

8. Table: products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Question:

Label stock status:

- 0 → 'Out of Stock'
- 1-5 → 'Low Stock'
- 5 → 'In Stock'

Product-Id	Stock-qty	stock-status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock.
5	3	Low stock

Expected Output Columns:

- product_id
- stock_qty
- stock_status

Product-Id	Stock-qty	stock-status
2	0	Out of stock
1	5	Low stock.
5	3	Low stock.

9. Table: **classes**

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Question:

Classify by size:

- $\geq 25 \rightarrow$ 'Large'
- $10-24 \rightarrow$ 'Medium'
- $< 10 \rightarrow$ 'Small'

Expected Output Columns:

- `subject`
- `enrolled_students`
- `class_size_category`

Subject	Enrolled - Students	Class-Size-Category
English	25	Large
Math	30	Large
Science	15	Medium
History	20	Medium
Art	5	Small

10. Table: payments

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

Question:

Apply discount flag:

- If `payment_method = 'Cash'` and `amount ≥ 200` → 'Eligible for Discount'
- Otherwise → 'Not Eligible'

Expected Output Columns:

- `payment_id`
- `payment_method`
- `amount`
- `discount_eligibility`

Payment-ID	Payment-method	Amount	Discount-eligibility.
1	Card	50.00	Not-Eligible
2	Cash	200.00	Eligible For Discount
3	Card	150.00	Not-Eligible
4	PayPal	75.00	Not-Eligible
5	Cash	300.00	Eligible For Discount